

Technical Document #24: Database Systems - Comprehensive Overview

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Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Sharding distributes data across multiple database servers. It improves performance by parallelizing query processing.

Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Database monitoring tracks performance metrics and identifies bottlenecks.
- Relational databases organize data into tables with rows and columns.
- Database migration tools manage schema changes and data transformations.